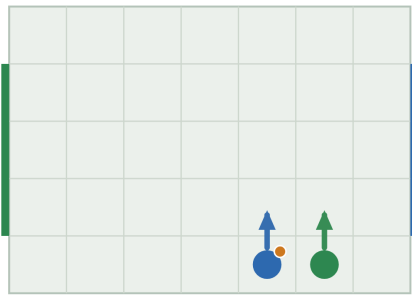


Player positions and the 4×4 Q matrix

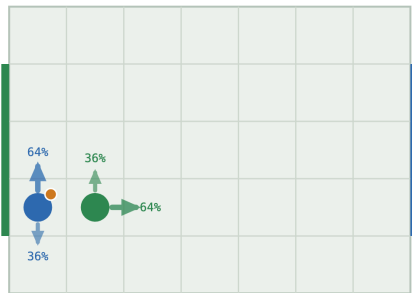
At the research meeting the professor drew a stage game by hand: a small grid of cells, each marked, with arrows chasing each other around the grid to show that no cell is a stable outcome. This is that sketch, generated from the exact solver instead of drawn free-hand, paired with a picture of where the two players actually are, and the exact, complete 4×4 {U, D, L, R} Q matrix underneath every case — always the full grid, never reduced, per the professor's own correction that the matrix should stay 4×4 and that the matrix itself, not an entropy number, is the actual output.

A pure state -- one safe cell, nothing to guess



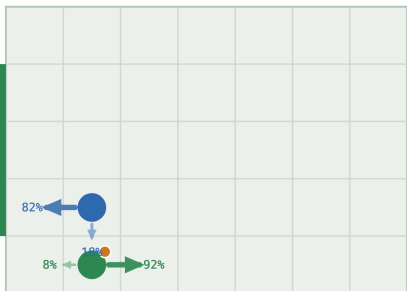
pure equilibrium · $V = +0.234$

The typical mix -- which goal row to head for



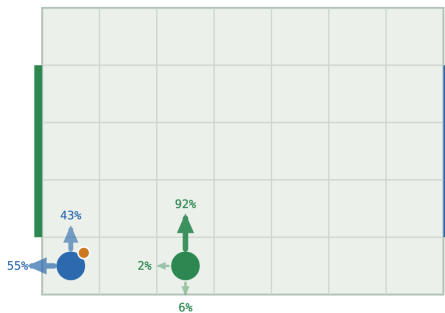
mixed equilibrium · $V = +0.094$

The meeting's example -- indifferent between L and R



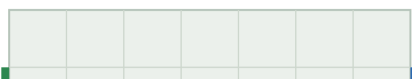
mixed equilibrium · $V = -0.206$

A genuine 3-action mix, not a 2-cycle

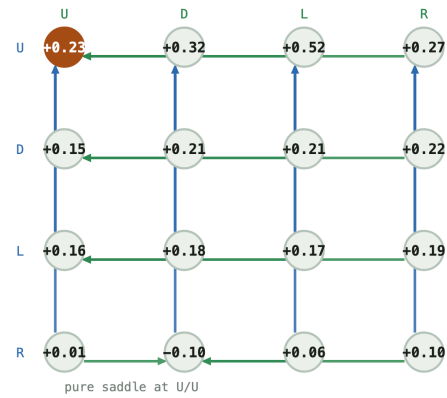


mixed equilibrium · $V = +0.085$

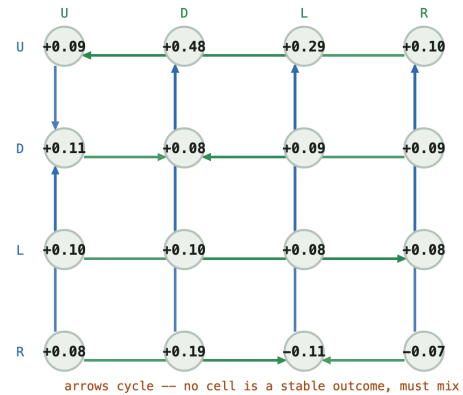
A near-pure hedge -- where rounding would lie



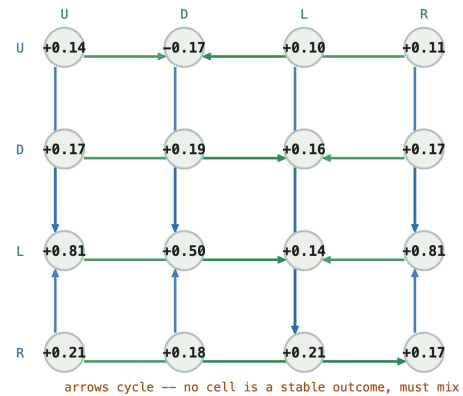
Q matrix -- (4, 0, 5, 0, 0)



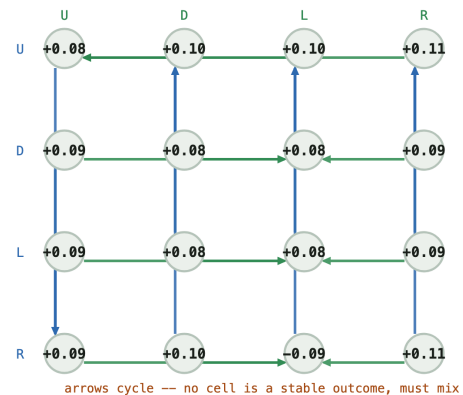
Q matrix -- (0, 1, 1, 1, 0)



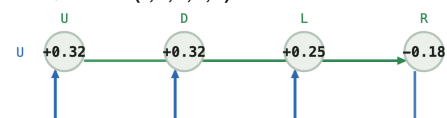
Q matrix -- (1, 1, 1, 0, 1)

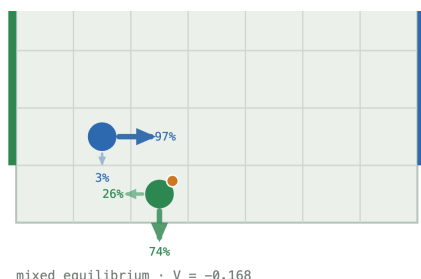


Q matrix -- (0, 0, 2, 0, 0)

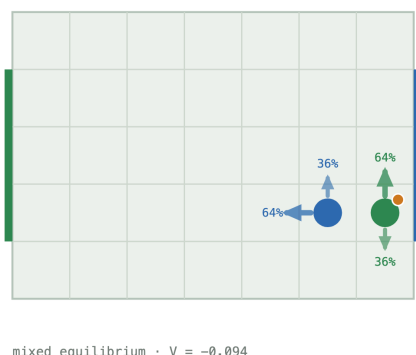


Q matrix -- (1, 1, 2, 0, 1)

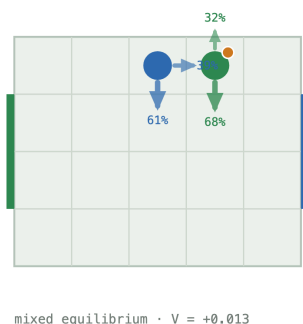




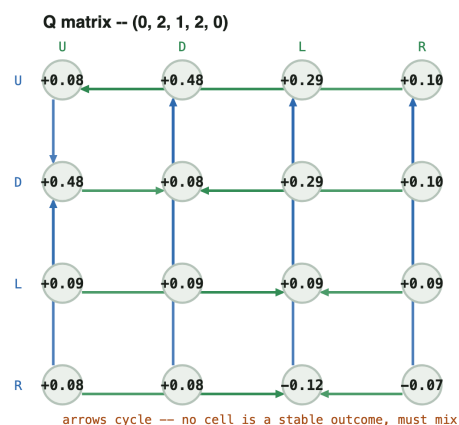
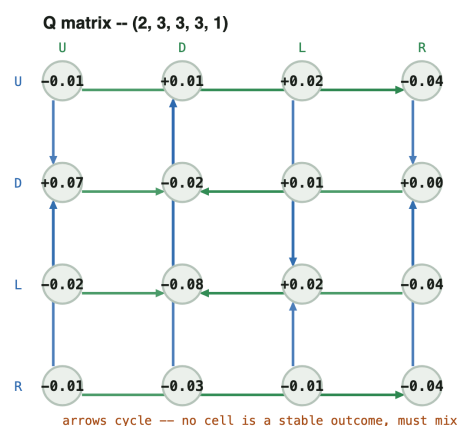
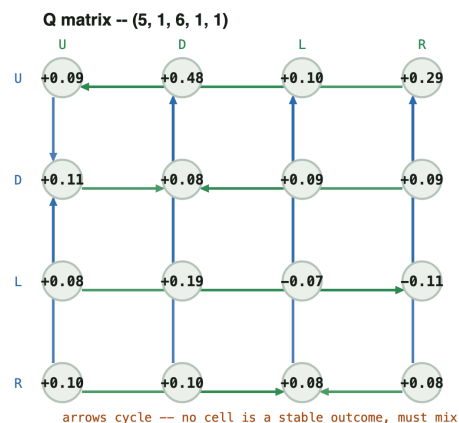
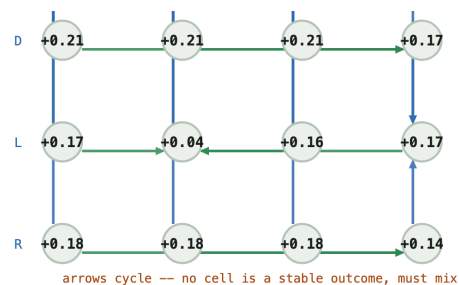
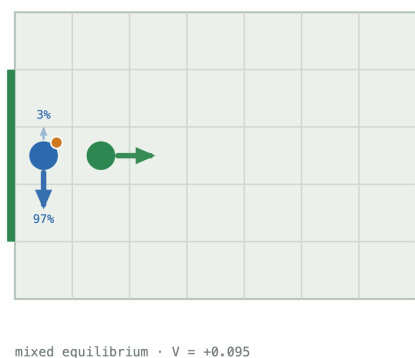
The typical mix, mirrored -- same shape, value negated



This project's own tackle rule -- a different mechanism, same duel



The asymmetric mix -- the carrier still guesses, the defender doesn't



All eight cases: board position (left) and the full 4×4 Q matrix redrawn as a node-and-arrow graph (right). A lit node with no outgoing arrow is a pure saddle; a closed loop of arrows means no cell is safe.

Reading the graph

Each node is one cell of the complete 4×4 Q matrix (row = the carrier's action, col = the defender's action, always reoriented so the printed payoff is the carrier's, regardless of which player id actually carries the ball). A blue arrow points from a cell to the cell in the same column with the higher payoff — the carrier's own reason to switch rows. A green arrow points from a cell to the cell in the same row with the lower payoff — the defender's reason to switch columns, since the defender minimises. Every case shows all 16 cells, so the shapes are directly comparable.

Case 1 (4, 0, 5, 0, 0)

A pure state, for contrast. Support (1,1) — a strict saddle at U/U. The carrier plays U always; the defender answers U always; neither has any reason to deviate.

	U	D	L	R
U	0.234	0.317	0.523	0.272
D	0.151	0.211	0.211	0.222
L	0.161	0.182	0.172	0.190
R	0.012	-0.095	0.058	0.098

Case 2 (0, 1, 1, 1, 0)

The typical mix. Support (2,2), gap 0.0136. The carrier mixes U 63.5% / D 36.5%; the defender answers U 36.5% / R 63.5%. This is the common shape: the carrier is deciding which goal row to head for, and the defender is guessing which one. **90 of the 94 mixed states on the canonical board cross a vertical pair exactly like this one.**

	U	D	L	R
U	0.086	0.478	0.286	0.099
D	0.109	0.076	0.085	0.086
L	0.103	0.103	0.085	0.084
R	0.082	0.187	-0.108	-0.071

Case 3 (1, 1, 1, 0, 1)

The meeting's own example.

"when we move right and when we move left, there's an equal chance of me winning. That's why I am indifferent between the two."

Support (2,2), gap 0.0445. The carrier's entire live option set is exactly $\{L, R\}$ — no vertical option survives at all, so this is the plain case the meeting asked for: two actions genuinely equally good against the defender's own mix (D 18% / L 82%). **This is the rarer shape: only 4 of 94 mixed states cross a purely horizontal pair instead of a vertical one.**

	U	D	L	R
U	0.139	-0.173	0.099	0.110
D	0.171	0.185	0.157	0.171
L	0.810	0.500	0.141	0.810
R	0.211	0.181	0.211	0.167

Case 4 (0, 0, 2, 0, 0)

A genuine 3-action mix. Support (3,3), gap 0.0047, entropy 1.109 bits — the richest mix on this page. Six cells from goal, as far as this board allows, with the defender close enough to threaten three different rows at once: no single row is safely better than the other two. Requested explicitly at the meeting: **the matrix itself, not entropy, is the point.**

	U	D	L	R
U	0.084	0.095	0.103	0.110
D	0.086	0.076	0.076	0.086
L	0.086	0.076	0.076	0.086
R	0.088	0.095	-0.088	0.110

Case 5 (1, 1, 2, 0, 1)

A near-pure hedge, where rounding would lie. Support (2,2) again, but entropy only 0.169 bits — the carrier plays R 97.5%, D 2.5%. Reading 0.975 as 1.0 would be the rounding mistake the meeting flagged: the gap (0.0043) is real and certified, twelve times smaller than a 0.1 rounding grid would resolve.

	U	D	L	R
U	0.317	0.317	0.247	-0.178
D	0.211	0.211	0.211	0.167
L	0.171	0.045	0.157	0.171
R	0.182	0.182	0.182	0.135

Case 6 (5, 1, 6, 1, 1)

The typical mix, mirrored. The board's left-right / role-swap symmetry maps Case 2 exactly onto this state. The values negate to machine precision:

```
V(case 2) = +0.094235
V(mirror) = -0.094235
sum       = -1.39e-17 (zero, to floating-point noise)
```

Direct evidence that Case 2's shape is not a one-off: the identical mix reappears, exactly mirrored, wherever the same relative configuration recurs.

	U	D	L	R
U	0.086	0.478	0.099	0.286
D	0.109	0.076	0.086	0.085
L	0.082	0.187	-0.071	-0.108
R	0.103	0.103	0.084	0.085

Case 7

(2, 3, 3, 3, 1), 5×4 board

This project's own tackle rule. Every other case here gets its coin flip from Littman's random move order. This one is solved under the tackle rule (defender commits to a challenge that wins the ball with fixed probability or bounces off) instead — a mechanism with nothing to do with move order at all. **It produces the same kind of duel** (support (2,2), gap 0.0223): the carrier mixes D 60.7% / R 39.3%, the defender mixes U 32.1% / D 67.9%. Different cause, same structure.

	U	D	L	R
U	-0.012	0.005	0.016	-0.042
D	0.072	-0.022	0.009	0.000
L	-0.019	-0.080	0.019	-0.042
R	-0.007	-0.026	-0.006	-0.039

Case 8

(0, 2, 1, 2, 0)

The asymmetric mix. Support (2, 1) — gap 0.0095. The odd one out: the carrier's equilibrium still mixes two actions (U 2.6% / D 97.4%), but the defender's equilibrium is a single fixed move (R, 100%). Every other case on this page is a symmetric duel — both players genuinely guessing. Here only one side is.

	U	D	L	R
U	0.085	0.478	0.291	0.095
D	0.478	0.085	0.291	0.095
L	0.093	0.093	0.086	0.093
R	0.082	0.082	-0.122	-0.072

Why the carrier still needs two actions against a defender who never varies: against the defender's pure R, column R reads U 0.095, D 0.095, L 0.093, R -0.072 — **U and D are exactly tied**, both strictly better than L or R. Nothing in the payoffs favours one over the other, so any split of U/D is an equally valid best reply; the solver's LP reports one particular split (2.6% / 97.4%), not a probability forced by the game the way Case 3's 18% / 82% is. This is the meeting's LP-edge-case question made concrete: **a tie between two rows can look identical, in the printed policy, to a genuine forced mix, but it is a different phenomenon** — no best-reply cycle is involved, and either pure U or pure D alone would also be a valid equilibrium reply to the defender's fixed R.

The mathematics behind all eight

A mixed equilibrium is exactly the strategy pair where every action in a player's support earns the **same expected payoff** against the opponent's own mix — that equality is what "indifferent" means in Case 3, and it is also why the best-response graph cycles with no resting point: if one action in the support paid strictly more, that player would raise its weight on it, which is precisely the condition an equilibrium rules out. A best-reply cycle and mutual indifference are the same fact seen from two sides, not two different

explanations. Case 8 shows the boundary of that story: indifference alone (a tie against a fixed opponent) can produce the same *symptom* — a fractional policy — without a cycle anywhere in the matrix.

Reproduce with `python scripts/positions.py (make positions)` — writes `figures/gallery/positions.svg` and prints every matrix and policy shown here. Full project: github.com/Charith-Reddy-Pareddy/soccer-markov-nash.